

# Dhruv Sureka

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## Education

### Georgia Institute of Technology

*Bachelor of Science in Computer Science — Concentration: ML Systems*

Aug. 2023 – May 2026

Atlanta, Georgia

## Experience

### Barclays

*Incoming Software Engineer*

May 2026

New York, NY

- Incoming engineer building ML models and infrastructure powering data-driven decision-making across investment banking and capital markets.

### Social Echelon

*Founder & Lead Developer*

Apr. 2025 – Present

Philadelphia, PA

- Created a digital-creator marketplace that utilizes a proprietary matching engine to connect brands with digital creators, facilitating over **\$15K** in automated partnership deals in its first phase.
- Engineered an agentic backend orchestration with **5 distributed microservices**, managing complex asynchronous tasks like real-time engagement tracking and automated brand pitching.
- Optimized platform performance to handle 10,000+ daily operations with sub-200ms response times, ensuring a seamless, low-latency experience for creators managing high-growth accounts.

### Georgia Tech Research

*Undergraduate Researcher*

Aug. 2024 – Present

Atlanta, GA

- Developed a real-time **Natural Language Processing** system to detect and translate domain-specific jargon into plain text, enabling seamless communication across cross-functional teams.
- Optimized inference time perf. for BERT by implementing **INT8 quantization** and **TensorRT**, achieving a 2x speedup and 75% memory reduction for real-time deployment.
- Fine-tuned Transformer** models to handle live jargon-to-text conversion, focusing on maintaining contextual accuracy while minimizing processing overhead.

### BreachRx

*Software Engineering Intern*

May 2024 – Aug. 2024

Philadelphia, PA

- Achieved **20%** more leads and boosted platform activity by **64%** for a cybersecurity startup by leveraging **AWS Lambdas**, **ChromaDB**, and semantic search, streamlining secure data retrieval and enhancing user engagement.
- Improved query accuracy and relevance by integrating OpenAI vector embeddings with a **Retrieval-Augmented Generation** model, enabling rapid summarization of complex legal information and strengthening the platform's cybersecurity compliance insights.
- Accelerated deployment speed and scalability using **Node.js**, **GraphQL**, and **AWS cloud** within an agile workflow, ensuring a robust, secure, and seamless user experience.

### Teaching Assistant

*Georgia Tech - CS 1332*

Jan. 2024 – Apr. 2024

Atlanta, GA

- Coordinated office hours, HWs, and exams for 300 students in Data Structures & Algorithms (Java) course.

## Projects

### High-Performance Vision Transformer | CUDA, PyTorch, PyTorch Lightning

Dec. 2025

- Developed a high-perf. ViT for Tiny ImageNet (200 classes) using PyTorch, achieving **3.4x inference speedup** through custom **CUDA kernels** that fuse bias addition and GELU activation, optimized with vectorized memory access (float4) and shared memory caching on H100 GPUs.
- Implemented distributed training pipeline on Georgia Tech's PACE ICE **HPC cluster** using PyTorch Lightning, Hydra, and Weights & Biases, with **SLURM** job scheduling and optimized data staging for distributed training on H100 GPUs.

## Technical Skills

**Languages:** Python, Java, C++, SQL, TypeScript

**ML Systems:** CUDA, Triton, TensorRT, vLLM, Flash Attention, ONNX

**ML Frameworks:** PyTorch, NumPy, SLURM, Distributed Training (DeepSpeed/FSDP)

**Optimization Techniques:** KV Cache, Continuous Batching, Speculative Decoding, MoE

**Architectures:** Transformers (Decoder-only), ViT, CNNs, Diffusion Models

**Infrastructure:** AWS, Docker, Kubernetes, Git, CI/CD, React, Node.js